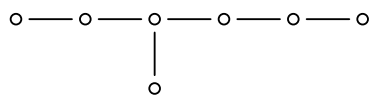
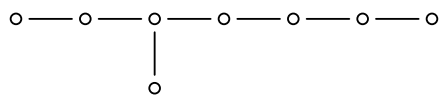
- E_7



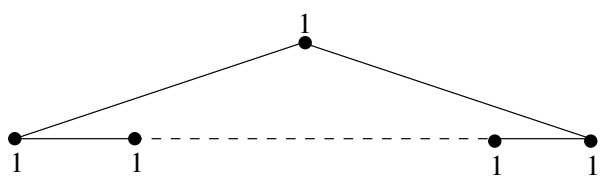
- E_8



(a) Compute the determinant of A where $\Gamma = A_n, D_n$. (Use the row decomposition rule, and write down a recursive equation for it.) Deduce by Sylvester criterion that A_n, D_n are Dynkin diagrams.[1]

(b) Compute the determinants of A for E_6, E_7, E_8 (use row decomposition and reduce to (a)). Show they are Dynkin diagrams.

(c) Show that if Γ is a Dynkin diagram, it cannot have cycles. For this, show that $\det(A) = 0$ for a graph Γ below:



(Show that the sum of rows is 0.) Thus Γ has to be a tree.

(d) Show that if Γ is a Dynkin diagram, it cannot have vertices with four or more incoming edges and that Γ can have no more than one vertex with three incoming edges. For this, show that $\det(A) = 0$ for a graph Γ below:

(e) Show that $\det(A) = 0$ for all graphs Γ below:

[1] The Sylvester criterion says that a symmetric bilinear form $(\ ,\ )$ on $\mathbb{R}^n$ is positive definite if and only if for any $k \leq n$, $\det_{1 \leq i,j \leq k}(e_i, e_j) > 0$.